Name: Jaikaran Singh Gill
Roll no: 8025320042

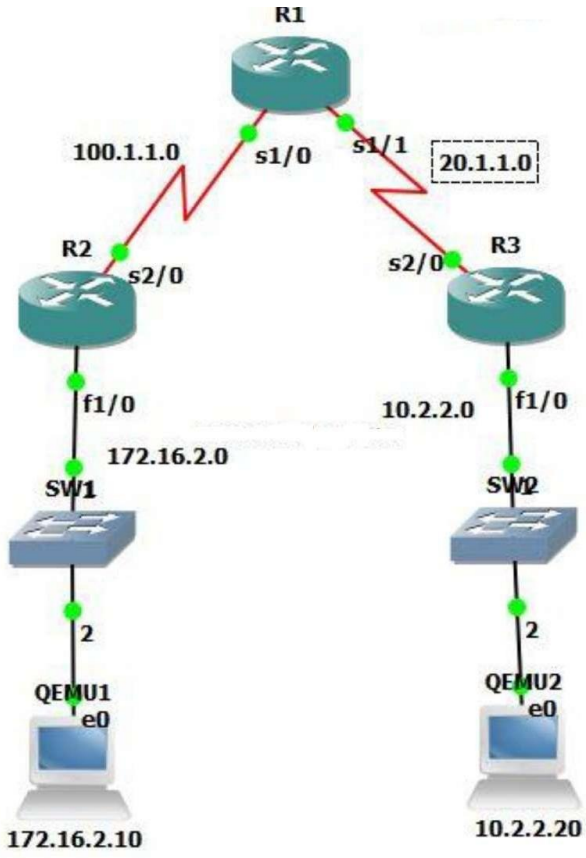
Assignment 6: Solution
Routing using RIP Version 2 (Cisco Packet Tracer)

Objective

To configure and verify routing using RIP Version 2 protocol in a network.

Problem Statement

Design and implement the given network topology in Cisco Packet Tracer with three routers (R1, R2, R3), two LANs, and PCs using the provided IP addressing scheme.

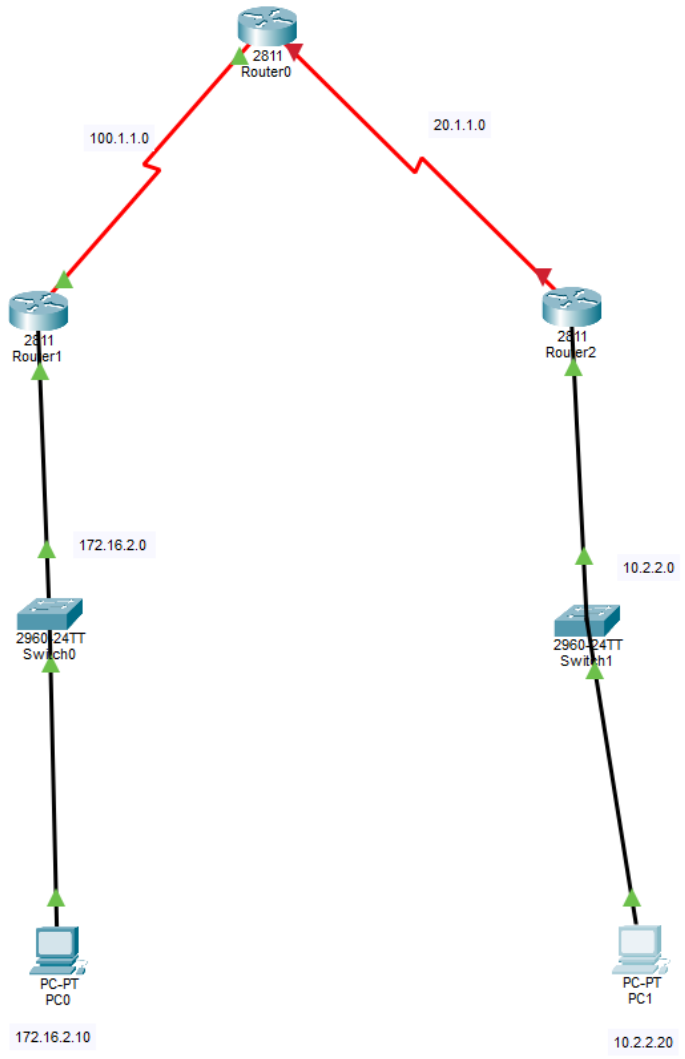


Network Details

Device	Interface	IP Address	Subnet Mask
R1	S0/0/0	100.1.1.2	255.255.255.0
R1	S0/0/1	20.1.1.1	255.255.255.0
R2	S0/0/0	100.1.1.1	255.255.255.0
R2	G0/0	172.16.2.1	255.255.0.0
R3	S0/0/1	20.1.1.2	255.255.255.0
R3	G0/0	10.2.2.1	255.255.255.0
PC1	NIC	172.16.2.10	255.255.0.0
PC2	NIC	10.2.2.20	255.255.255.0

Tasks

- Create the network topology in Cisco Packet Tracer.
- Configure IP addresses on all router interfaces and PCs.
- Enable RIP Version 2 routing on all routers.
- Advertise all directly connected networks using RIP.
- Disable auto-summary in RIP configuration.
- Verify routing tables using 'show ip route'.
- Test connectivity by pinging PC2 from PC1 and between routers.

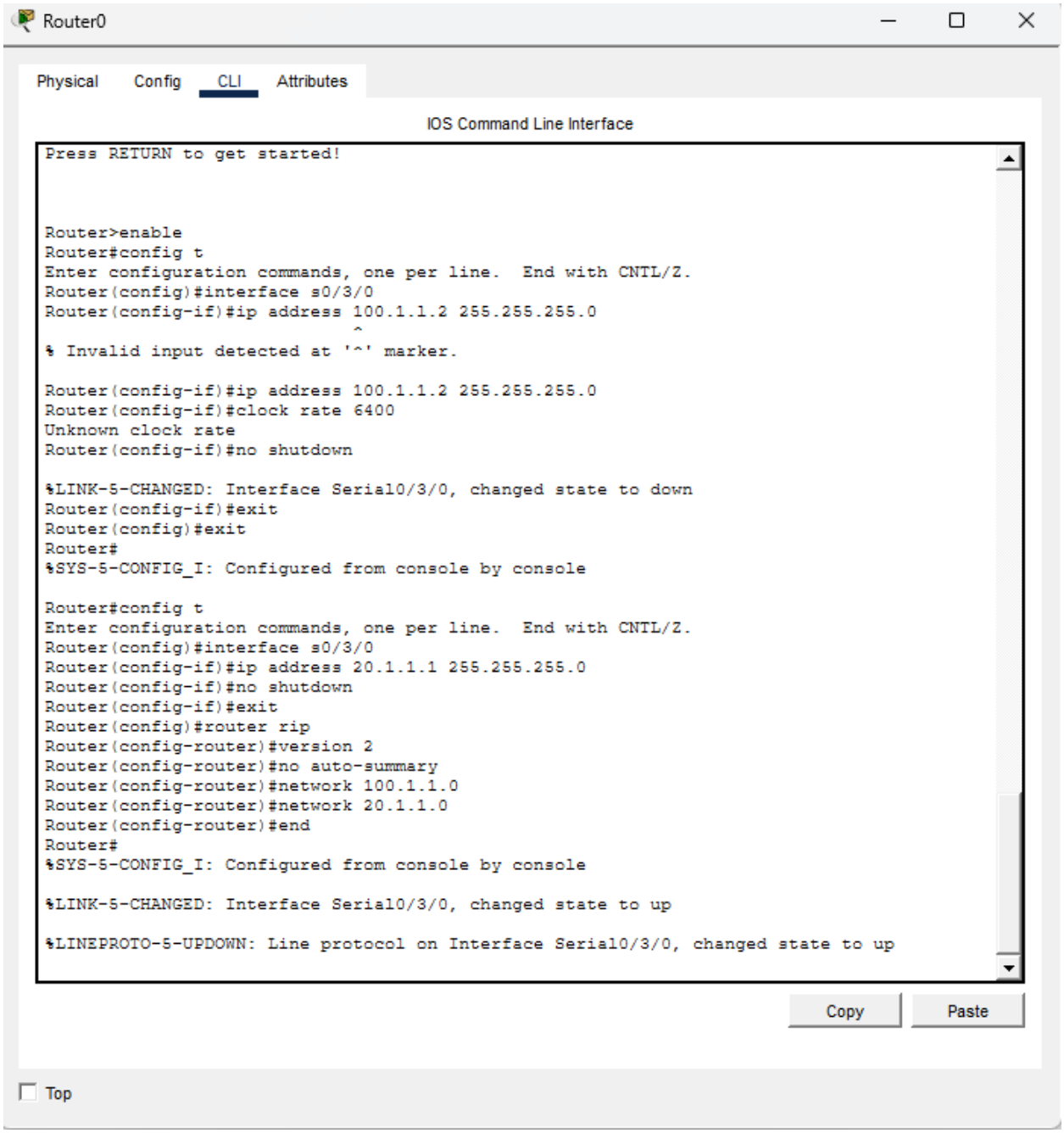


Router Configurations

R1 Configuration:

```
enable
conf t
interface s0/0/0
ip address 100.1.1.2 255.255.255.0
clock rate 64000
no shutdown
interface s0/0/1
ip address 20.1.1.1 255.255.255.0
no shutdown
router rip
version 2
no auto-summary
```

network 100.1.1.0
network 20.1.1.0
end

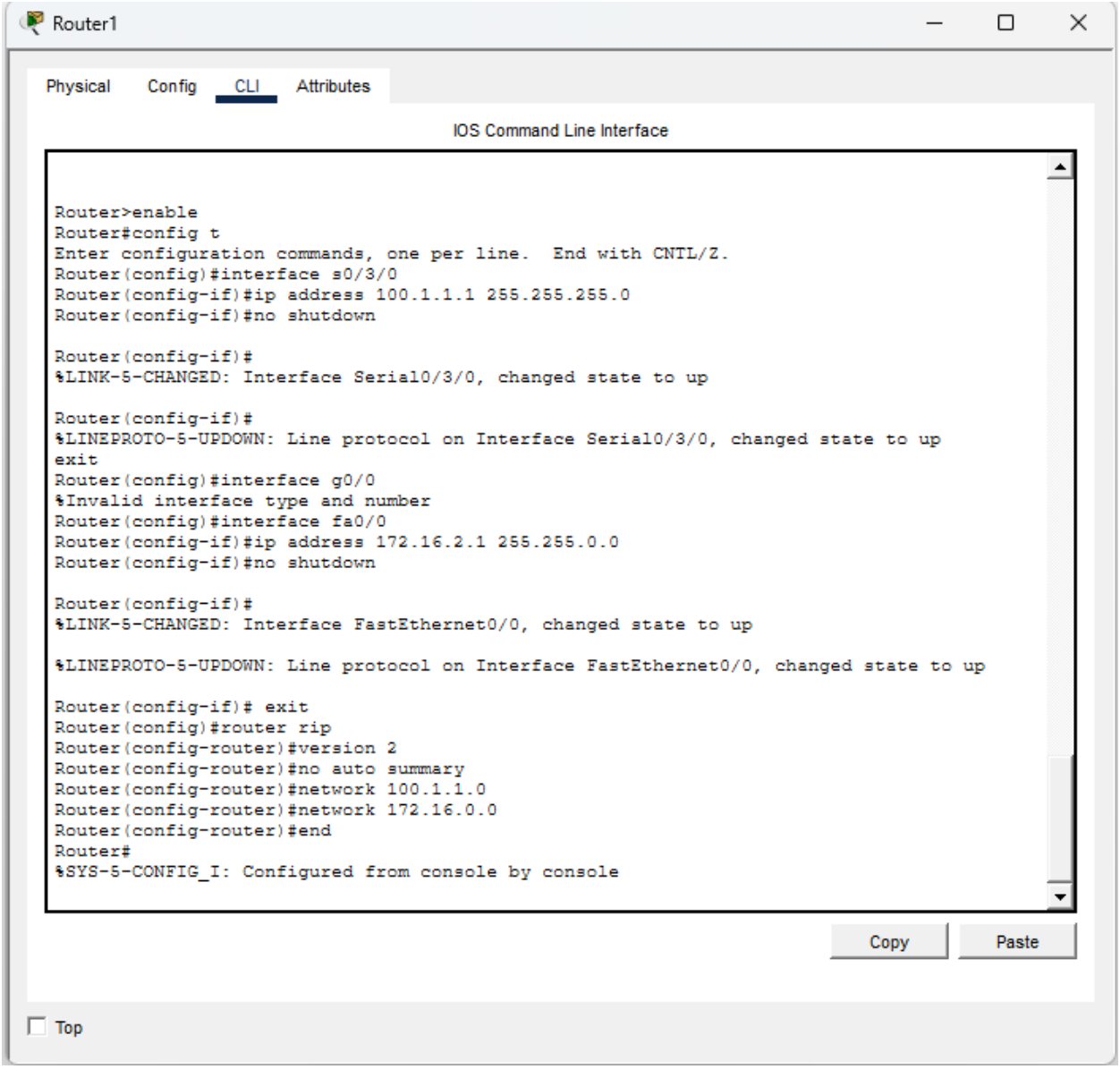



```
Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface s0/3/0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
                             ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#clock rate 6400
Unknown clock rate
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface s0/3/0
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 20.1.1.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

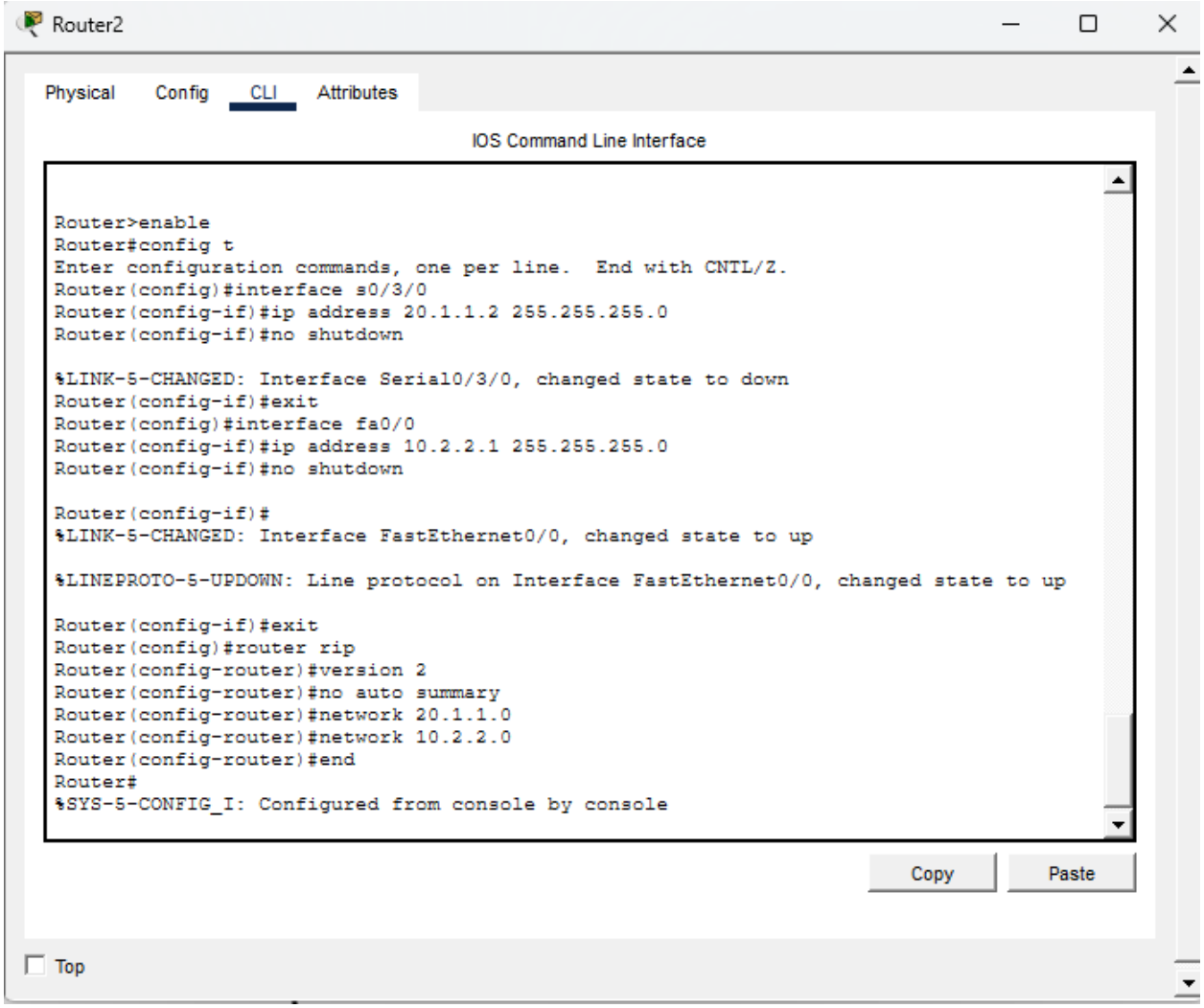
R2 Configuration:

enable
conf t
interface s0/0/0
ip address 100.1.1.1 255.255.255.0
no shutdown
interface g0/0
ip address 172.16.2.1 255.255.0.0
no shutdown

router rip
version 2
no auto-summary
network 100.1.1.0
network 172.16.0.0
end



R3 Configuration:

enable
conf t
interface s0/0/1
ip address 20.1.1.2 255.255.255.0
no shutdown
interface g0/0
ip address 10.2.2.1 255.255.255.0
no shutdown
router rip

version 2
no auto-summary
network 20.1.1.0
network 10.2.2.0
end